

XOR parity tree

Parity counts whether the number of ones is even or odd

Fold, depth, even and odd

- Four-bit one-zero-one-zero has two ones, so reduce XOR equals zero
- Level one pairs one xor zero and one xor zero to get one-one
- Tree depth for four bits is two; chain depth is three
- Even parity bit equals reduce XOR; odd parity is the inverse
- Flip any single bit and parity toggles
- An odd leftover at a tree level promotes unchanged to the next row

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

XOR parity tree

Fold bits with a **balanced XOR reduction tree** — same parity as a linear chain, but $\text{depth} \approx \log_2 N$ instead of $N-1$.

Starter example: 4-bit 1010 — reduce XOR = 0 (even ones). Balanced tree depth 2 vs chain depth 3.

Load starter example

Challenges (1/22)

Quiz: reduce. Parity of ones equals which reduction? Answer: XOR

Your answer

Show hintCheckNextIdle

quiz-reducequiz-assocquiz-logquiz-chainstarterdepth-4step-oncelevel1-starterfull-rootcompareeven-podd-preset-8depth-8all1-4toggleexplainquiz-promotehex-loadflip-parityquiz-samefull-path

Core ideas

Reduce XOR

Parity of ones = $b_0 \oplus b_1 \oplus \dots$ (odd $\rightarrow 1$).

Tree

Pairwise XOR; odd leftover promotes — depth $\lceil \log_2 N \rceil$.

Chain

Serial fold needs $N-1$ XOR delays on the critical path.

XOR parity tree starter

learn_digital — xor parity tree

Workbook practice

- In the workbook track, reduce-xor the bits one-one-zero-one by hand
- Draw a balanced tree for eight bits and state tree depth versus chain depth
- For reduce XOR equals one, give even and odd parity bits
- Name one pitfall: building a long XOR chain when a tree would meet timing

Pitfalls to watch

- Do not confuse even parity with odd parity, one is the complement of the other
- Tree and chain must agree on the final bit, but not on delay
- And remember: the browser lab is literacy
- Real links still need generator polynomials, multi-bit ECC

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, sketch one four-bit tree and label each level
- When you are ready, take the short quiz, then continue to tri-state and bus